

Java开发环境配置

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1. 下载JDK

- JDK下载地址: <https://www.oracle.com/java/technologies/downloads/#jdk21-windows>

JDK 21 JDK 17 GraalVM for JDK 21 GraalVM for JDK 17

JDK Development Kit 21.0.1 downloads

JDK 21 binaries are free to use in production and free to redistribute, at no cost, under the Oracle No-Fee Terms and Conditions (NFTC).

JDK 21 will receive updates under the NFTC, until September 2026, a year after the release of the next LTS. Subsequent JDK 21 updates will be licensed under the Java SE OTN License (OTN) and production use beyond the limited free grants of the OTN license will require a fee.

Linux macOS **Windows**

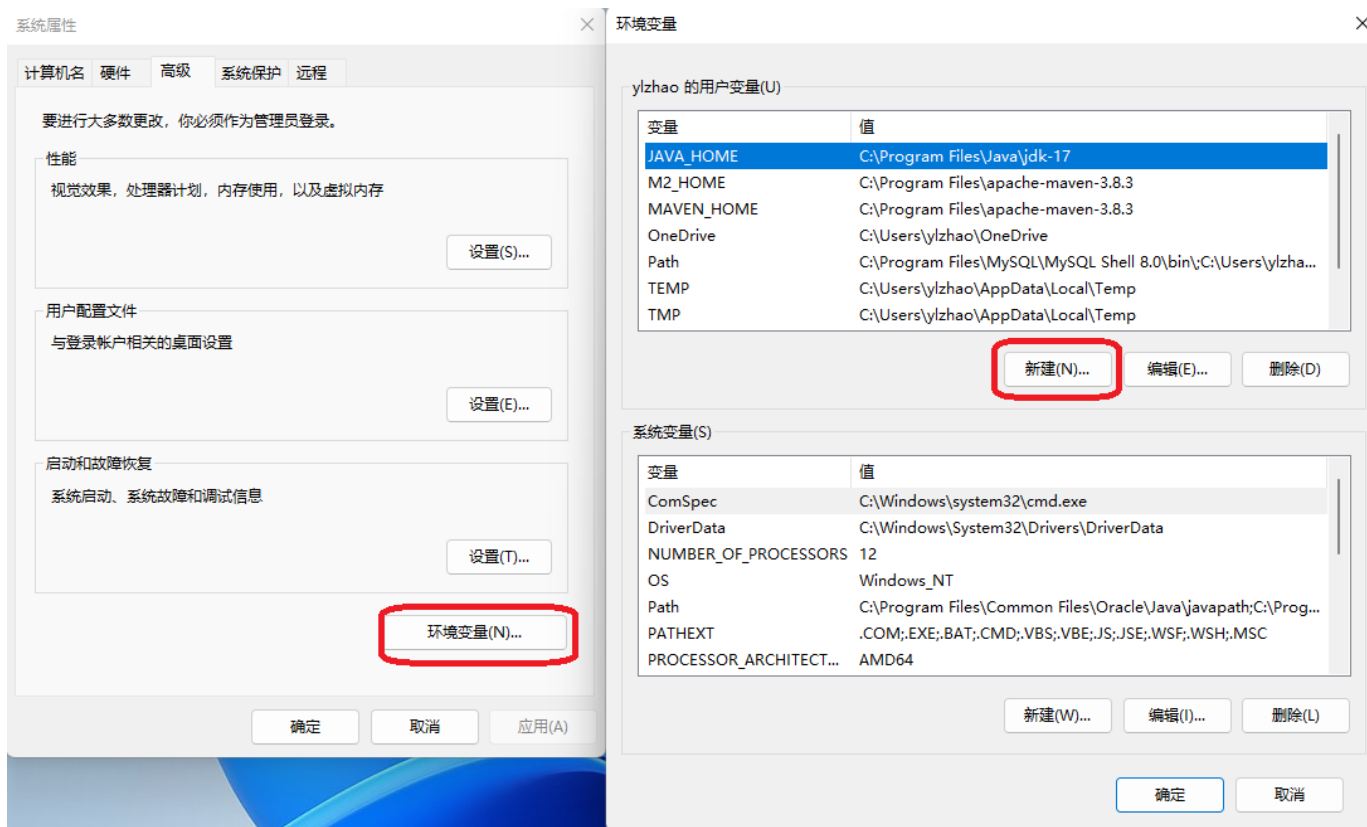
Product/file description	File size	Download
x64 Compressed Archive	185.39 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.zip (sha256)
x64 Installer	163.82 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.exe (sha256)
x64 MSI Installer	162.60 MB	https://download.oracle.com/java/21/latest/jdk-21_windows-x64_bin.msi (sha256)

2. 安装JDK

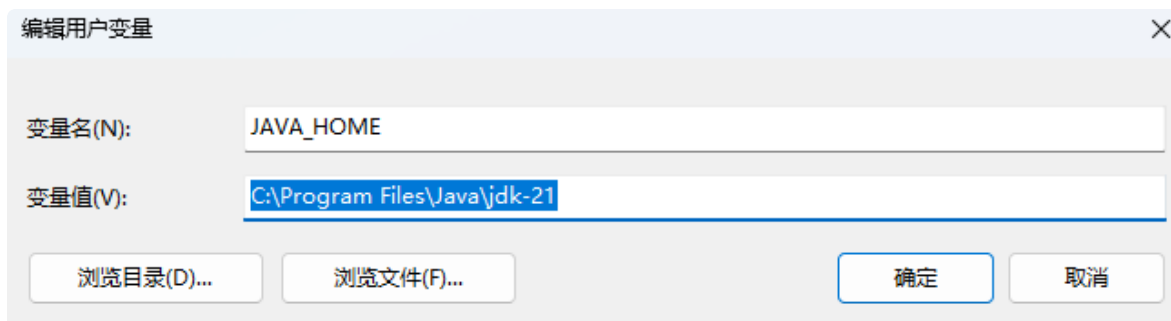
- 双击"jdk-21_windows-x64_bin.msi"文件进行安装即可，安装路径选择默认的路径。

3. JDK环境变量配置

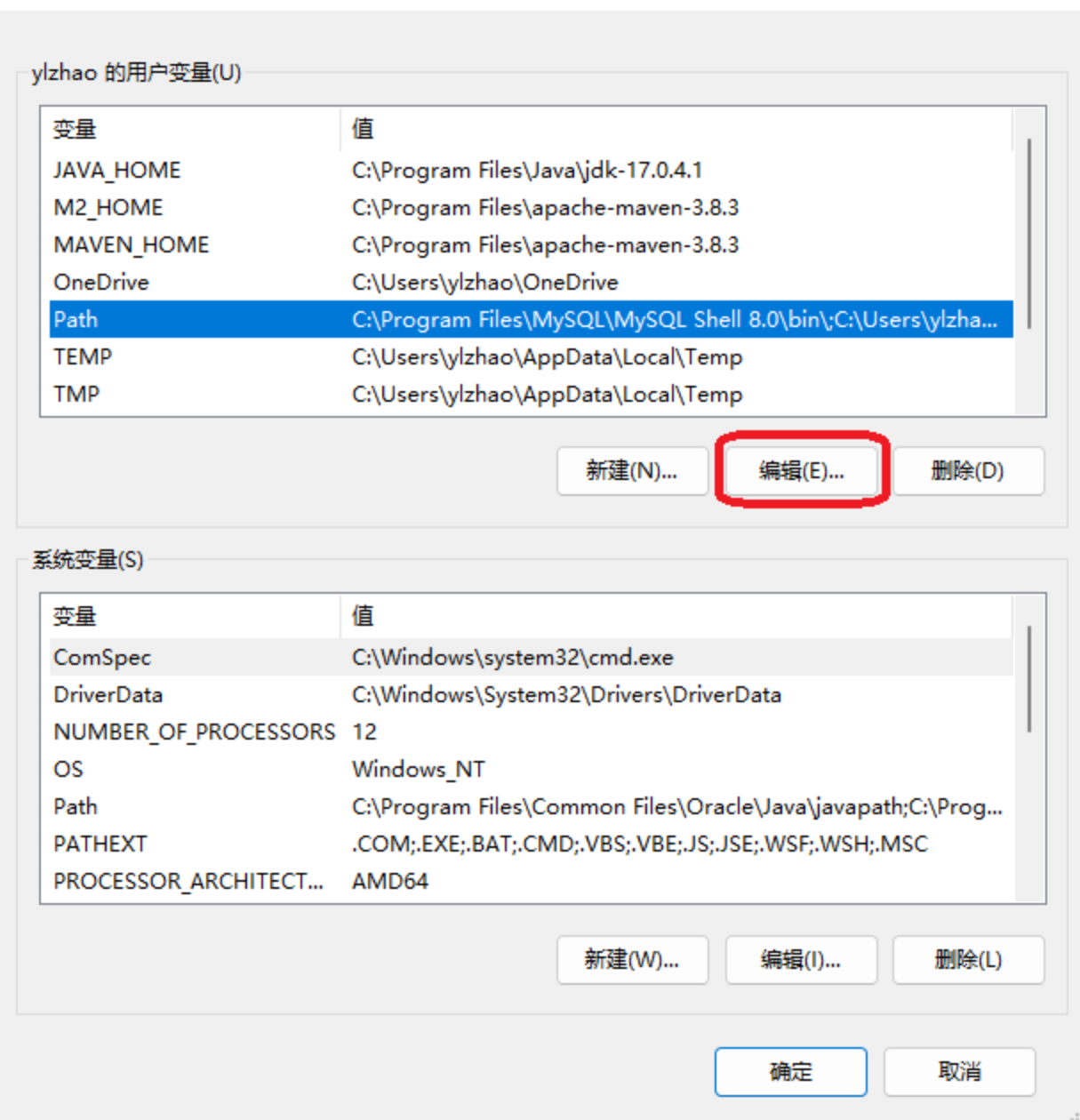
- 打开系统属性窗口，选择环境变量，然后选择新建按钮。



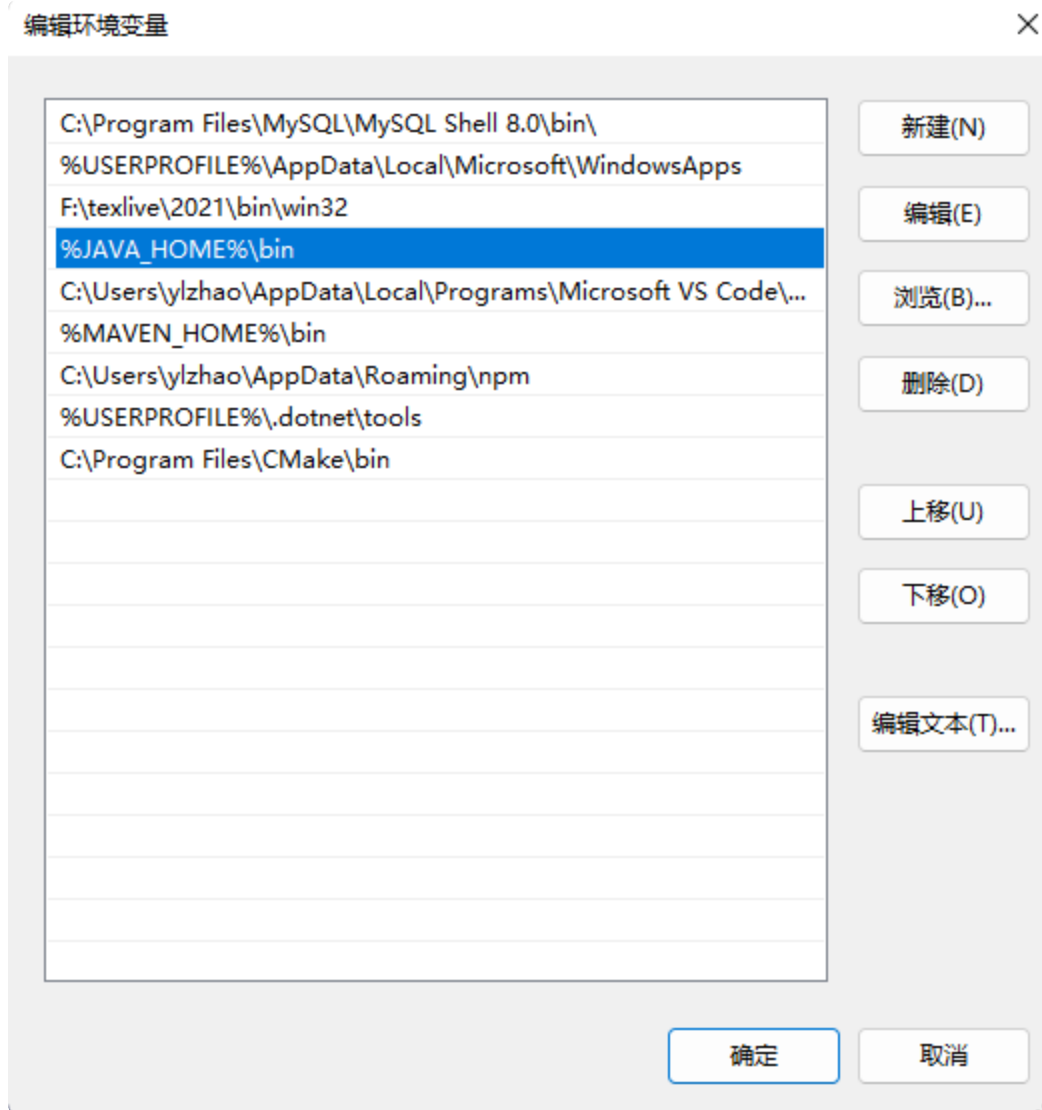
- 新建环境变量Java_HOME



- 编辑Path环境变量



- 在Path环境变量中添加新的一项"%JAVA_HOME%\bin"



- 添加完成后依次点击确定按钮。

4. JDK安装验证

- 打开终端，输入命令"java -version"，如下图所示，表示JDK安装正确。

```
命令提示符
Microsoft Windows [版本 10.0.22631.2861]
(c) Microsoft Corporation。保留所有权利。

C:\Users\zyl>java -version
java version "21.0.1" 2023-10-17 LTS
Java(TM) SE Runtime Environment (build 21.0.1+12-LTS-29)
Java HotSpot(TM) 64-Bit Server VM (build 21.0.1+12-LTS-29, mixed mode, sharing)
```

5. 配置Maven

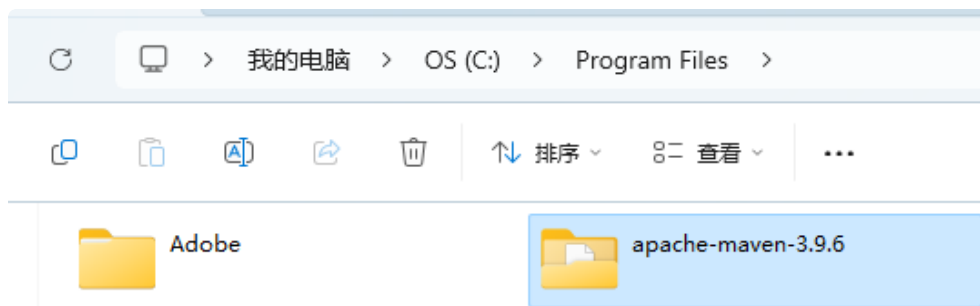
- Maven下载地址: <https://maven.apache.org/download.cgi>

Files

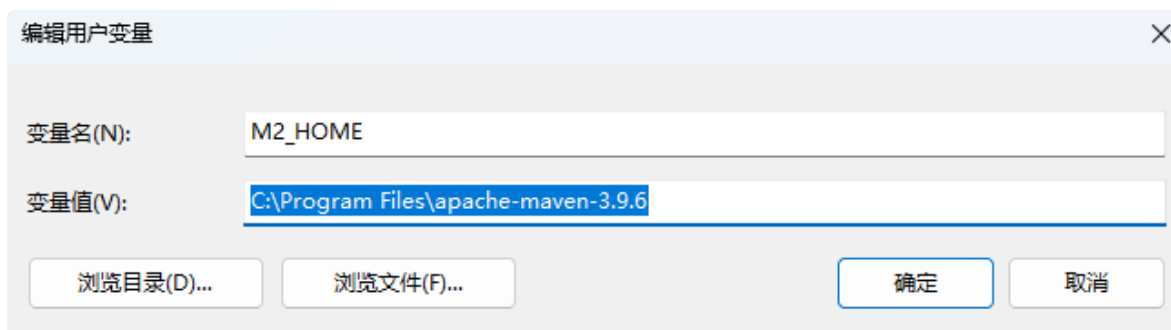
Maven is distributed in several formats for your convenience. Simply pick a ready-made binary distribution. In order to guard against corrupted downloads/installations, it is highly recommended to verify the

	Link
Binary tar.gz archive	apache-maven-3.9.6-bin.tar.gz
Binary zip archive	apache-maven-3.9.6-bin.zip
Source tar.gz archive	apache-maven-3.9.6-src.tar.gz
Source zip archive	apache-maven-3.9.6-src.zip

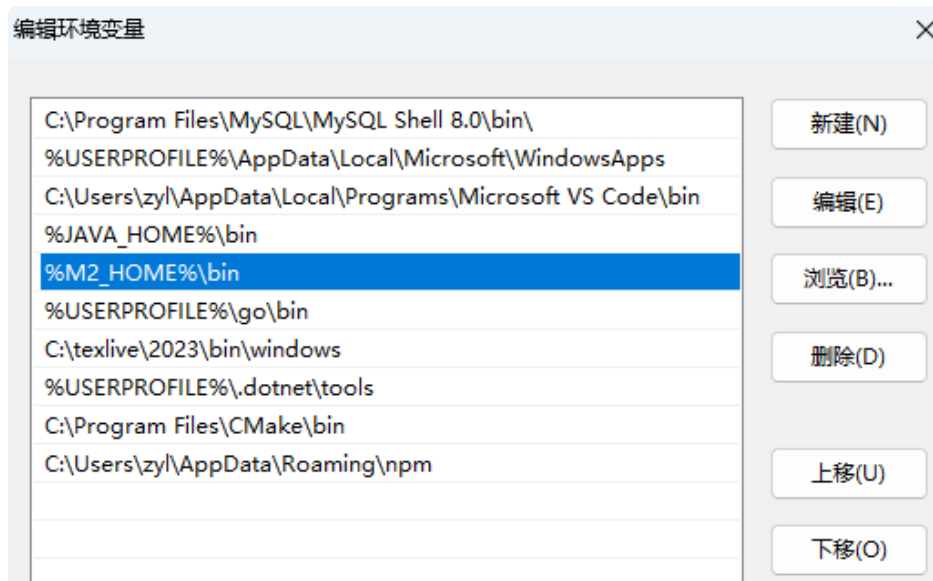
- 将下载好的文件解压到C:\Program Files目录中。



- 新建环境变量M2_HOME



- 在Path环境变量中添加新的一项"%M2_HOME%\bin"



6. Maven安装验证

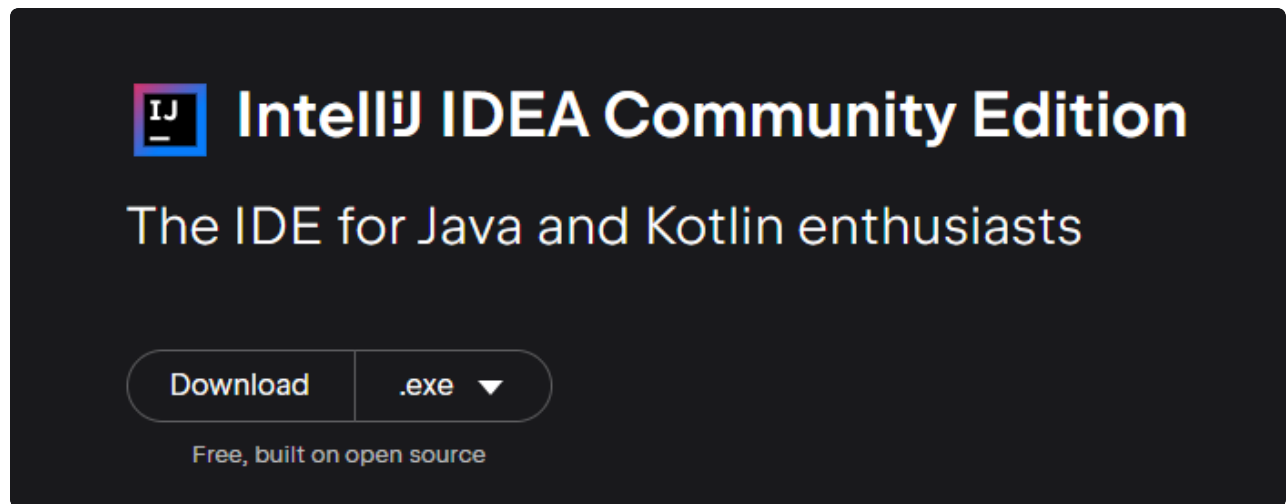
- 打开终端，输入命令"mvn -version"，如下图所示，表示Maven安装正确。

```
命令提示符
Microsoft Windows [版本 10.0.22631.2861]
(c) Microsoft Corporation。保留所有权利。

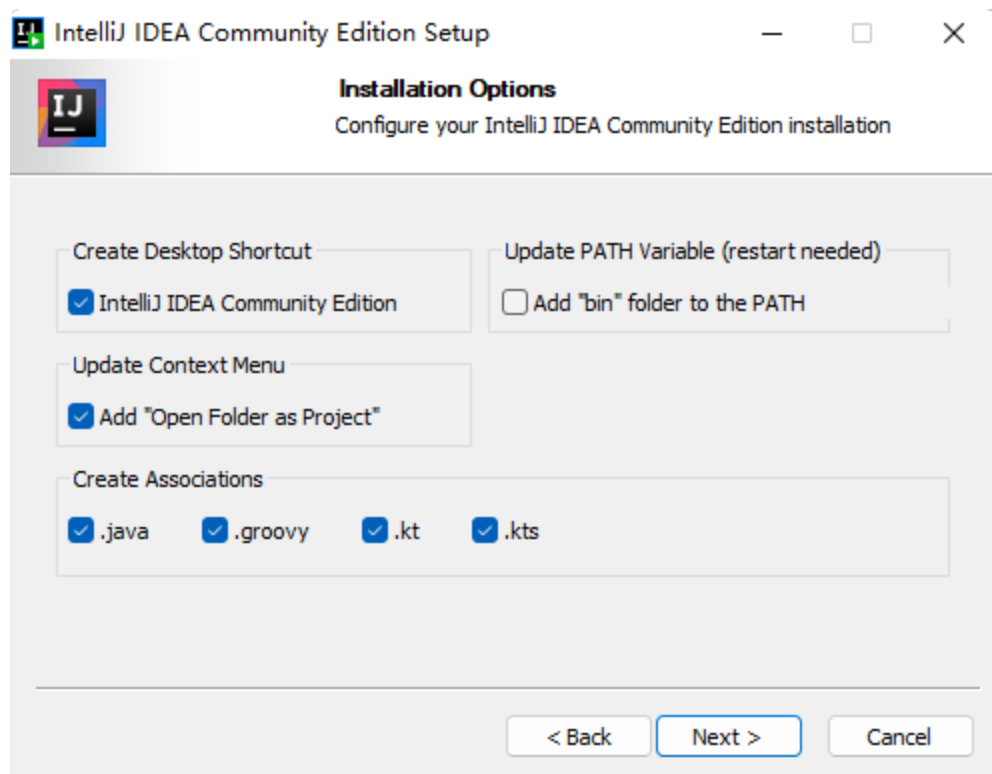
C:\Users\zyl>mvn -version
Apache Maven 3.9.6 (bc0240f3c744dd6b6ec2920b3cd08dcc295161ae)
Maven home: C:\Program Files\apache-maven-3.9.6
Java version: 21.0.1, vendor: Oracle Corporation, runtime: C:\Program Files\Java\jdk-21
Default locale: zh_CN, platform encoding: UTF-8
OS name: "windows 11", version: "10.0", arch: "amd64", family: "windows"
```

7. IntelliJ IDEA安装

- IntelliJ IDEA是Java语言的集成开发环境，依赖于JDK。
- IntelliJ IDEA下载地址：<https://www.jetbrains.com/idea/download/?section=windows>
- 请下载Community版本

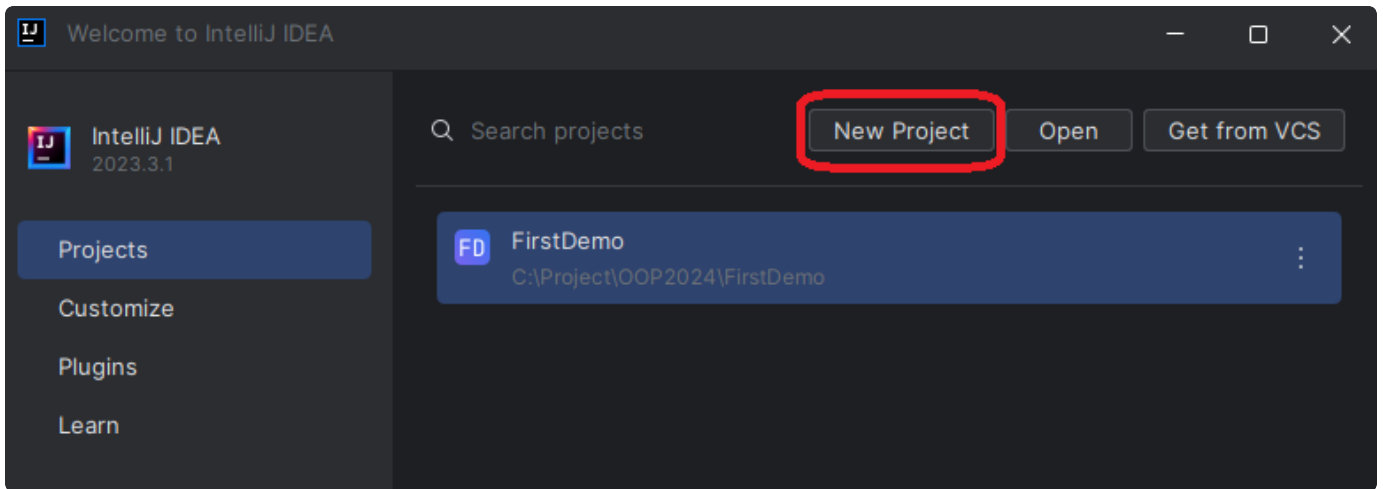


- 双击下载的文件进行安装，安装过程中注意选择：

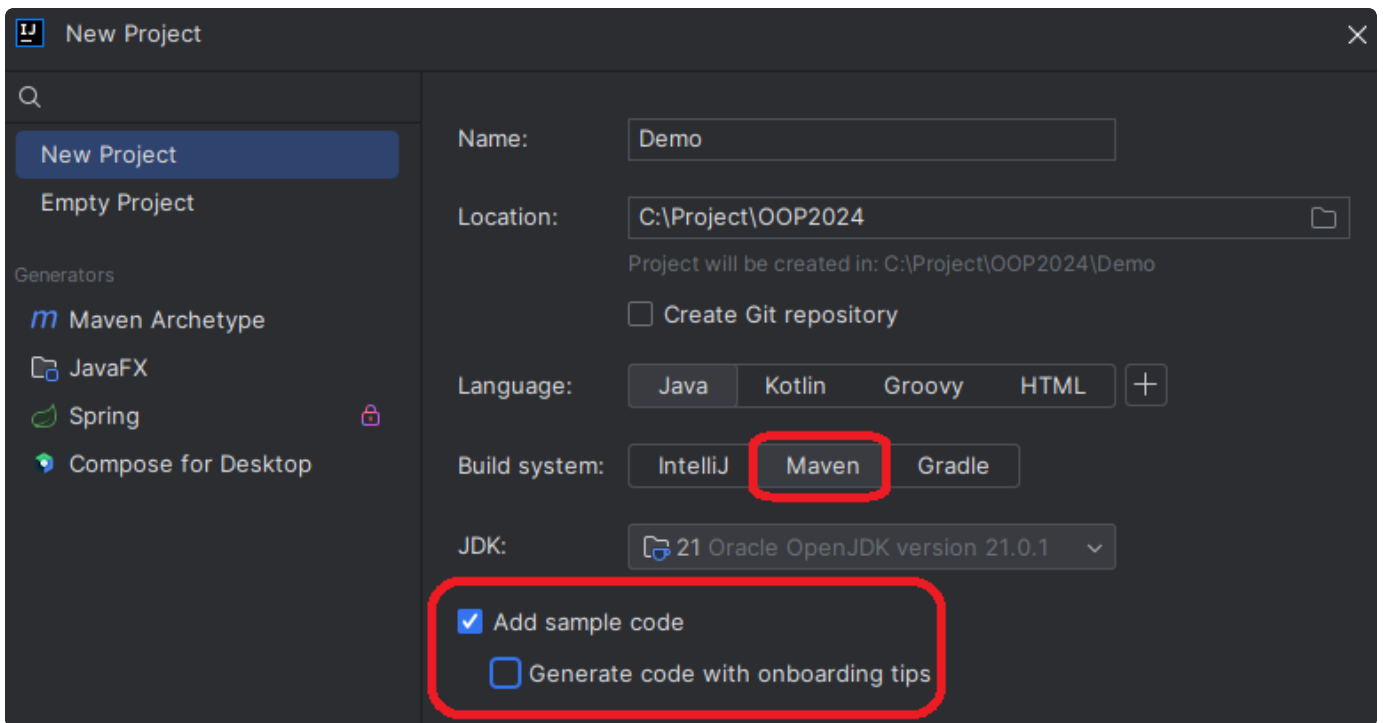


8. 使用IntelliJ IDEA创建Java项目

- 启动IntelliJ IDEA，选择"New Project"创建一个新的项目。



- 注意构建系统选择"Maven"。



- 项目创建完成后，默认 pom.xml 文件内容：

```
1  <?xml version="1.0" encoding="UTF-8"?>
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
3          xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4          xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5      <modelVersion>4.0.0</modelVersion>
6
7      <groupId>org.example</groupId>
8      <artifactId>Demo</artifactId>
9      <version>1.0-SNAPSHOT</version>
10
11     <properties>
12         <maven.compiler.source>21</maven.compiler.source>
13         <maven.compiler.target>21</maven.compiler.target>
14         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
15     </properties>
16
17 </project>
```

- 使用快捷键Ctrl+F9构建项目
- 使用快捷键Ctrl+Shift+F10运行项目。

9. 将项目打包为jar文件运行

- 修改pom.xml文件，添加<build>标签部分的内容。

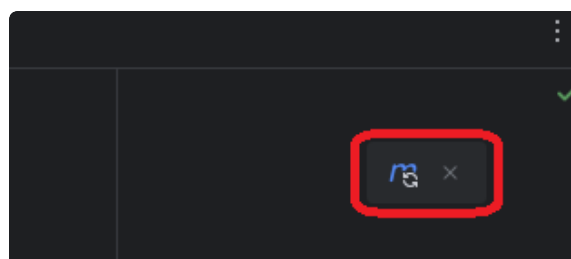
pom.xml
Java

```

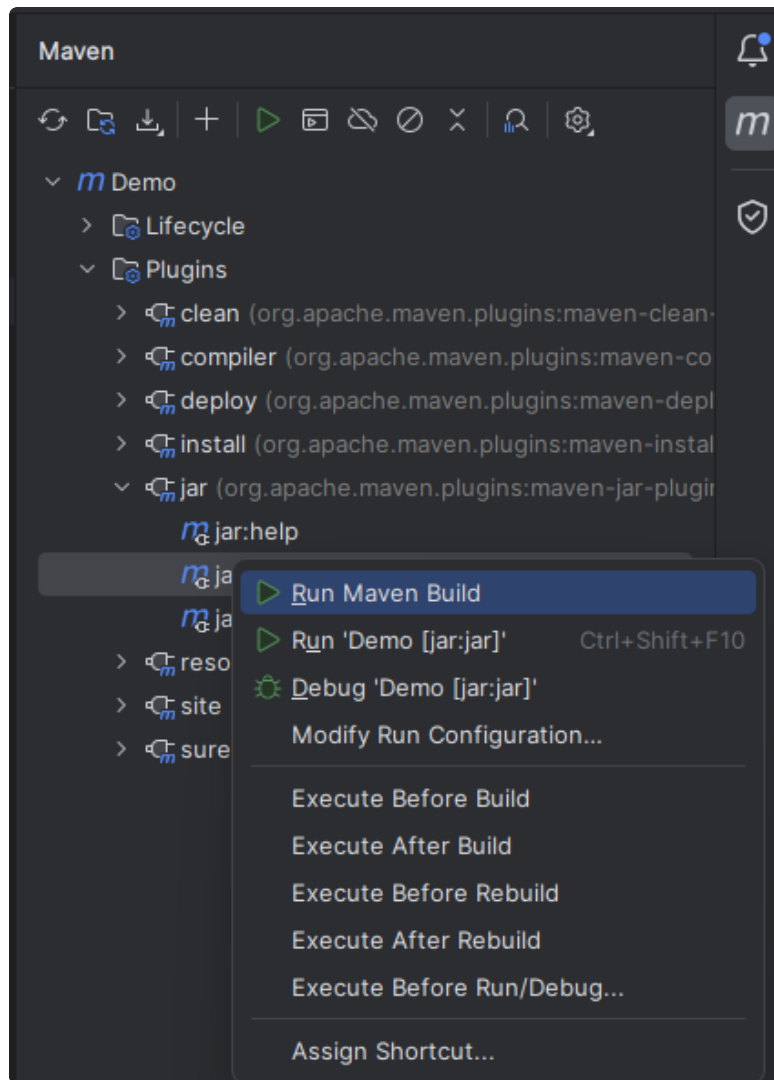
1  <?xml version="1.0" encoding="UTF-8"?>
2  <project xmlns="http://maven.apache.org/POM/4.0.0"
3      xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
4      xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/xsd/maven-4.0.0.xsd">
5      <modelVersion>4.0.0</modelVersion>
6
7      <groupId>org.example</groupId>
8      <artifactId>FirstDemo</artifactId>
9      <version>1.0-SNAPSHOT</version>
10
11     <properties>
12         <maven.compiler.source>21</maven.compiler.source>
13         <maven.compiler.target>21</maven.compiler.target>
14         <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
15     </properties>
16
17     <build>
18         <plugins>
19             <plugin>
20                 <groupId>org.apache.maven.plugins</groupId>
21                 <artifactId>maven-jar-plugin</artifactId>
22                 <version>3.3.0</version>
23                 <configuration>
24                     <archive>
25                         <manifest>
26                             <addClasspath>true</addClasspath>
27                             <mainClass>org.example.Main</mainClass>
28                         </manifest>
29                     </archive>
30                 </configuration>
31             </plugin>
32         </plugins>
33     </build>
34
35 </project>

```

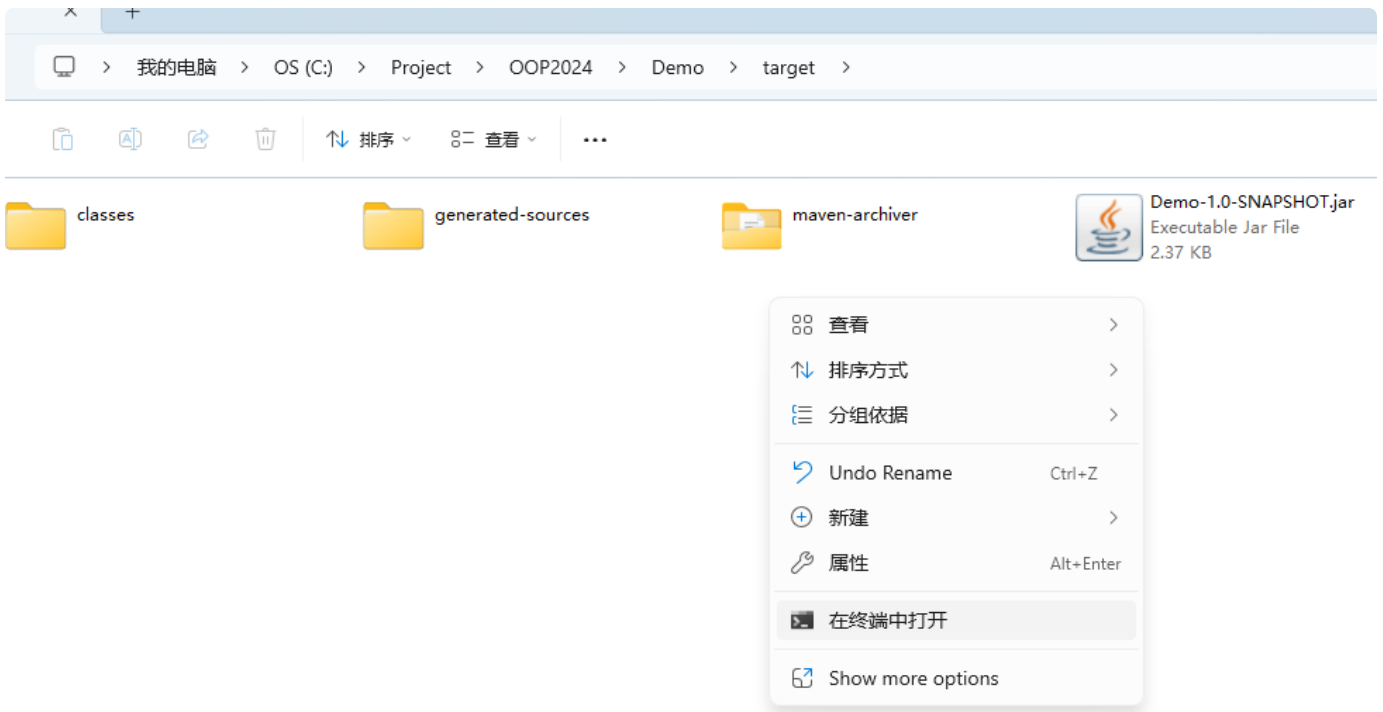
- 注意1: 修改pom.xml文件后, 点击窗口右上角的maven按钮, 使得修改生效。



- 注意2：这个时候Maven会从主仓库中下载所需要的包，因此电脑需要联网。
- 点击窗口右边的m图标，打开Maven窗口。在"jar:jar"选项上面单击鼠标右键，选择"Run Maven Build"。



- 在项目的"target"目录里面打开终端



- 在终端运行命令：java -jar Demo-1.0-SNAPSHOT.jar

```
命令提示符
Microsoft Windows [版本 10.0.22631.2861]
(c) Microsoft Corporation。保留所有权利。

C:\Project\OOP2024\Demo\target>java -jar Demo-1.0-SNAPSHOT.jar
Hello world!
i = 0
i = 1
i = 2
i = 3
i = 4
i = 5

C:\Project\OOP2024\Demo\target>
```

10. 使用Maven添加第三方包

- Maven仓库：<https://mvnrepository.com/>

11. 安装Node.js

- Node.js下载地址：<https://nodejs.org/en>

Node.js® is an open-source, cross-platform JavaScript runtime environment.

Download Node.js®

20.11.1 LTS

Recommended For Most Users

21.6.2 Current

Latest Features

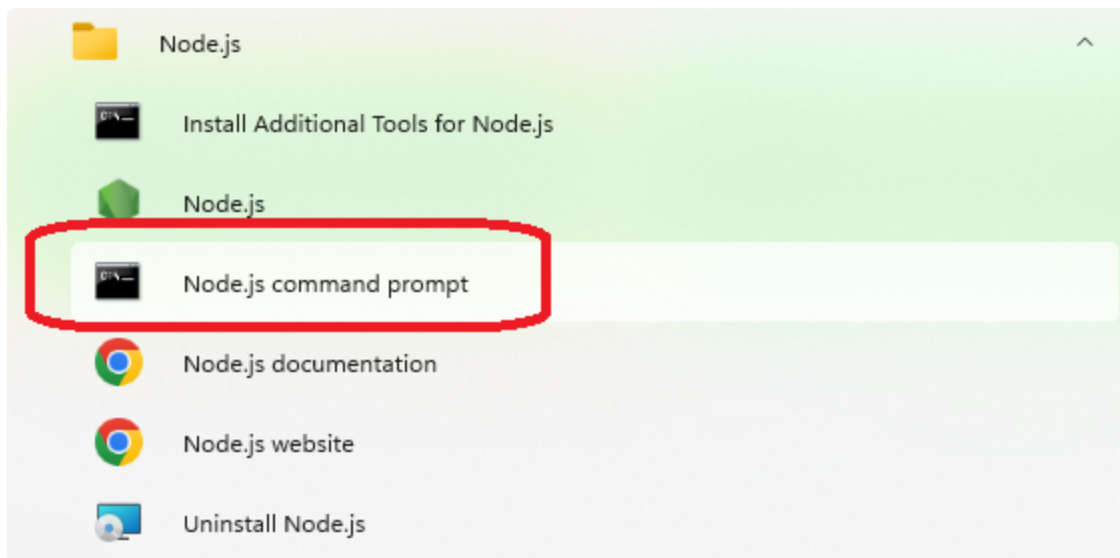
[Other Downloads](#) | [Changelog](#) | [API Docs](#) [Other Downloads](#) | [Changelog](#) | [API Docs](#)

For information about supported releases, see the [release schedule](#).

- 下载20这个版本
- 双击node-v20.11.1-x64.msi文件进行安装

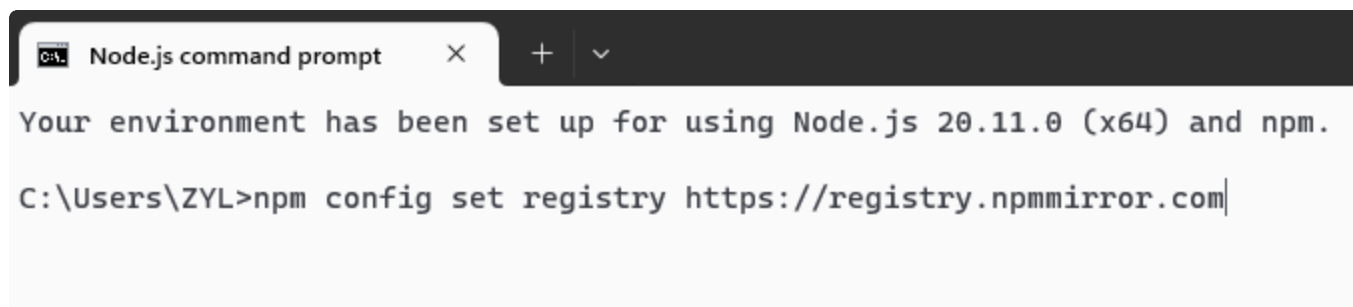
12. 配置Node.js镜像源

- 在开始菜单中单击Node.js command prompt选项



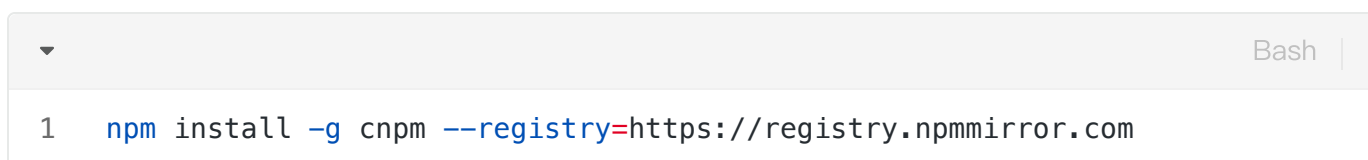
- 如下图所示，将npm设置淘宝镜像加速。

```
▼ Bash |  
1  npm config set registry https://registry.npmmirror.com
```

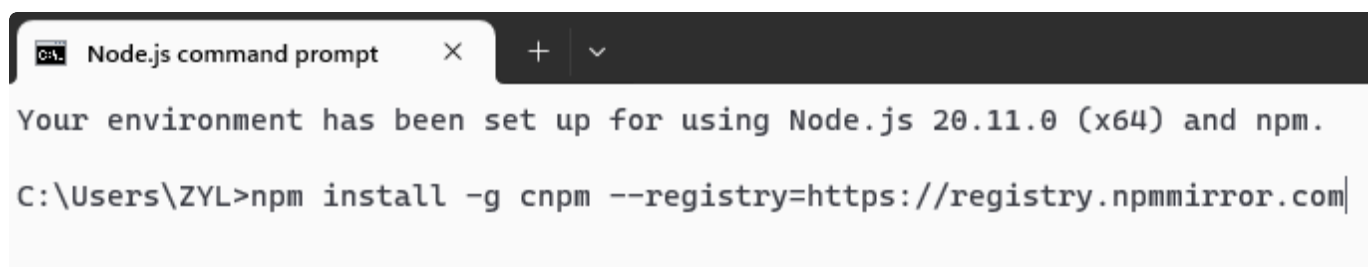


```
Node.js command prompt
Your environment has been set up for using Node.js 20.11.0 (x64) and npm.
C:\Users\ZYL>npm config set registry https://registry.npmmirror.com|
```

- 如下图所示安装cnpm。



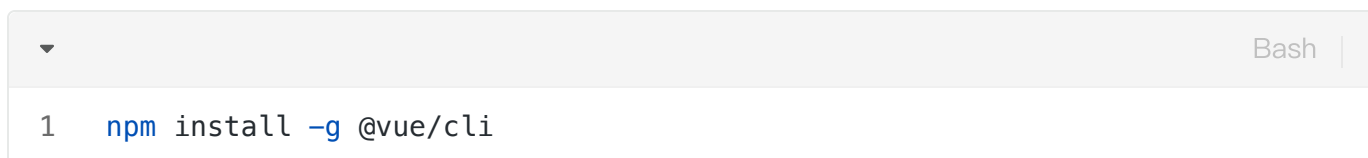
```
Bash
1 npm install -g cnpm --registry=https://registry.npmmirror.com
```



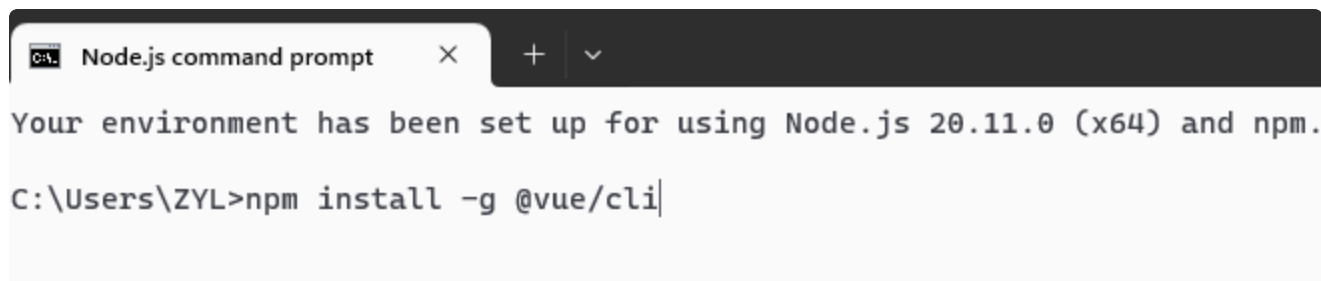
```
Node.js command prompt
Your environment has been set up for using Node.js 20.11.0 (x64) and npm.
C:\Users\ZYL>npm install -g cnpm --registry=https://registry.npmmirror.com|
```

13. 安装Vue 2.0

- 在开始菜单中单击Node.js command prompt选项
- 安装Vue 2.0参考: <https://cli.vuejs.org/zh/guide/installation.html>

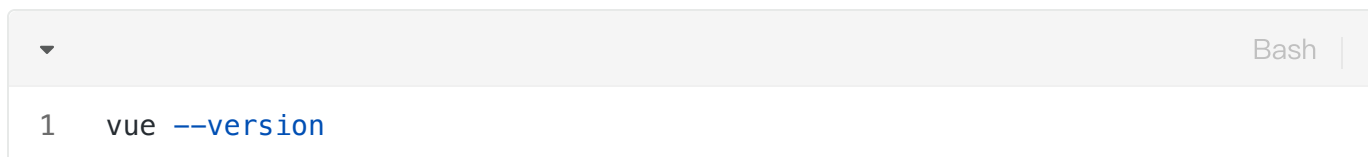


```
Bash
1 npm install -g @vue/cli
```



```
Node.js command prompt
Your environment has been set up for using Node.js 20.11.0 (x64) and npm.
C:\Users\ZYL>npm install -g @vue/cli|
```

- 检查Vue的版本



```
Bash
1 vue --version
```

C:\WINDOWS\system32\cmd. X

+ v

Your environment has been set up for using Node.js 20.11.0 (x64) and npm.

C:\Users\ZYL>vue --version

@vue/cli 5.0.8

C:\Users\ZYL>|